$$\begin{cases} x+2y+z=5 \\ 2x+3y+z=6 \\ 3x+y+2z=8 \end{cases}$$

$$\Downarrow$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 2 & 3 & 1 & 6 \\ 3 & 1 & 2 & 8 \end{array}\right]$$

Converter para 0
triangulo inferior
esquerdo

$$\Downarrow$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 2 & 3 & 1 & 6 \\ 3 & 1 & 2 & 8 \end{array}\right]$$

Verificar se a11=1

$$\Downarrow$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 2 & 3 & 1 & 6 \\ 3 & 1 & 2 & 8 \end{array}\right]$$

Converter L2 para 0

$$\Downarrow$$

$L_2 = L_2 - 2 \times L_1$

$2 - 2 \times 1 = 0$

$3 - 2 \times 2 = -1$

$1 - 2 \times 1 = -1$

$6 - 2 \times 5 = -4$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & -1 & -1 & -4 \\ 3 & 1 & 2 & 8 \end{array}\right]$$

$$\Downarrow$$

Converter L3 para 0

$L_3 = L_3 - 3 \times L_1$

$3 - 3 \times 1 = 0$

$1 - 3 \times 2 = -5$

$2 - 3 \times 1 = -1$

$8 - 3 \times 5 = -7$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & -1 & -1 & -4 \\ 0 & -5 & -1 & -7 \end{array}\right]$$

$$\Downarrow$$

$L_3 = L_3 - 5 \times L_2$

$-5 - 5 \times (-1) = -5 + 5 = 0$

$-1 - 5 \times (-1) = -1 + 5 = 4$

$-7 - 5 \times (-4) = -7 + 20 = 13$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & -1 & -1 & -4 \\ 0 & 0 & 4 & 13 \end{array}\right]$$

$$\begin{array}{cccc} x & y & z & \\ \end{array}$$
$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & -1 & -1 & -4 \\ 0 & 0 & 4 & 13 \end{array}\right]$$

R: $\boxed{x = \frac{7}{4} \quad y = \frac{3}{4} \quad z = \frac{13}{4}}$

$1x + 2y + 1z = 5$

$-1y - 1z = -4$

$4z = 13$

$\frac{\cancel{4}z}{\cancel{4}} = \frac{13}{4}$

$\boxed{z = \frac{13}{4}}$

$-1y - \cancel{\frac{13}{4}} = -4$

$\cancel{+\frac{13}{4}} \qquad +\frac{13}{4}$

$-1y = -4 + \frac{13}{4}$

$-1y = \frac{-16}{4} + \frac{13}{4}$

$\frac{\cancel{-1}y}{\cancel{-1}} = \frac{-3}{4}$

$\boxed{y = \frac{3}{4}}$

$1x + 2 \times \frac{3}{4} + \frac{13}{4} = 5$

$1x + \frac{6}{4} + \frac{13}{4} = 5$

$1x + \cancel{\frac{19}{4}} = 5$

$\cancel{-\frac{19}{4}} \qquad -\frac{19}{4}$

$1x = 5 - \frac{19}{4}$

$1x = \frac{20}{4} - \frac{19}{4}$

$\frac{\cancel{1}x}{\cancel{1}} = \frac{1}{4}$

$\boxed{x = \frac{1}{4}}$